Population Projections & Demographic Forecasting

Lecture 1: Population Projections & Cohort Component Method

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Presentations

- A little bit about myself …
- … and a little bit about you

Miscellaneous

- Course materials available on Moodle and on GitHub
- Please ask questions during the class or email me at any time (also after this week)
- Lectures consist of theoretical slides and small exercises in R to make the class more entertaining
- On top of projecting and forecasting, we will learn how to use animations to make nice dynamic visualizations, and how to use shiny apps
- Course should be propaedeutic to Adrian's Raftery course on "*Bayesian projection methodology*" in May

Assignment

- Group assignment, with three students for each group
- Five exercises (out of nine), which will be given after each class
- The exercises follow up on what we have touched during the class
- Please email me a single .pdf file containing your answers, as well as the R codes and data to generate the results. Ideally you can send me an R markdown document
- Deadline: Sunday 28th June

Course overview

- Lecture 1: Population Projections & Cohort Component Method
- Lecture 2: Matrix Projections I
- Lecture 3: Matrix Projections II
- Lecture 4: Demographic Forecasting & Lee-Carter

Pop. projections: motivations I

"*Population projection is probably the demographic technique that is most frequently requested by demography's clients*" (Preston et al. 2001)

- Governments: to anticipate demands for schools, roads, medical personnel, …
- Private businesses: potential size of their future market
- Public and private sector: future pension expenditures

⇒ more generally, they are the **base** for many other projections that are essential for social and economic planning, e.g. labour, education, health, housing, social security, energy, climate, transport, …

Pop. projections: motivations II

Demographers use population projections for a variety of needs:

- to analyse the implications of a certain set of demographic parameters on population size, composition and growth
- to help understand the determinants of population change and inform policy discussions
- to illustrate the implications of certain demographic characteristics (the model's *inputs*) on population parameters over time (the model's *outputs*)
- typical inputs: fertility, mortality and migration - although methodology can be extended to different demographic processes (marriage patterns, contraceptive use regimes, …)
- to identify realistic goals and targets for future development trends

Projections vs forecast I

"*Population projections are calculations which show the future development of a population when certain assumptions are made about the future course of fertility, mortality and migration. They are in general purely formal calculations, developing the implications of the assumptions that are made.*" (United Nations 1958)

- quality of projections is determined by their *internal* validity: do projections obey fundamental demographic accounting relations?
- can be used to answer "*what if*" questions, i.e. purely hypothetical scenarios corresponding to different assumptions. For example, what would happen to the population:
  - *if* mortality had not changed in the last ten years
  - *if* fertility had been 10% lower
  - *if* …

Projections vs forecasts II

"*A population forecast is a projection in which the assumptions are considered to yield a realistic picture of the probable future development of a population*" (United Nations 1958)

- quality of forecasts is determined by their *external* validity: are the predictions of subsequent events accurate?
- important feature of forecasts is their *uncertainty*
- stochastic forecasts are characterized by probabilistic distributions from which one can compute:
  - central (point) forecast
  - prediction interval, allowing to make probabilistic statement of forecast uncertainty

Projections vs forecasts III

In Nathan Keyfitz (1972) words: "*Population projections are correct beyond any testing against the subsequent population performance; in fact they can be incorrect only in the trivial sense that the author made an arithmetic error that prevents his final numbers from being consistent with his initial assumptions.*"

"*Yet in fact projections and forecasts are not easily distinguished. … A demographer makes a projection, and his reader uses it as a forecast.*"

$\Rightarrow$ in this course we will look at both projections and forecasts

Global population projections

- **United Nations** World Population Prospects (WPP) (web-link, visualization)
  - since 1946, published every two/three years
  - probabilistic projection model since 2015
  - very good track record of accurate projections
- **IIASA** Population and Human Capital Projections (web-link, visualization)
  - since 1970s, published less frequently
  - first global projections by age, sex and **educational attainment**
  - based on over 550 expert opinions
- **IHME** (paper, visualization)
  - more recent
  - based on an epidemiological approach to answer policy-oriented “what-if” questions
  - received some criticism from demographic community (see here)
- **US Census Bureau** (web-link, visualization)

Population projection methodology

A desirable methodology should:

- obey the demographic accounting relationships (balancing equation of population growth)
- be implementable with the available inputs
- provide the desired level of detail in the output:
  - only total population
  - disaggregated by age and/or sex
  - 1y or 5y intervals
  - including additional information (education, ethnicity, urban/rural, …)

Cohort-component method I

The cohort-component method (CCM):

- it can be traced back to Cannan (1895) and Westergaard (1908) (as reported by Smith and Keyfitz (1977) and de Gans (2003))
- nearly the only method used for population projections, almost universal consensus among social scientists
- employed by the United Nations for their WPP population estimates and projections, as well as by several statistical agencies (i.e. Federal Statistical Office of Germany, EUROSTAT, …)
- model's intuition: segment the population into different groups exposed to specific fertility, mortality and migration "risks", and compute the changes over time in each group
- typical groups: age and sex; can be extended to include race, nationality, location (regions, rural/urban), educational attainment, …

Cohort-component method (CCM) II

Three main reasons for the success of the CCM (Burch 2003):

- *completeness & simplicity*: complete account of size and age-sex composition of populations, based on relatively simple mathematics
- *explanation & prediction*: given inputs of fertility, mortality and migration, the CCM provides an exhaustive account of past and future population dynamics
- *guidance*: can be used to answer “what if” questions of population dynamics, eg. how demographic components could be used as policy levers

Cohort-component method III

- discrete-time model
- projection period divided into time intervals of the same length as the age intervals employed
- for each subgroup, three main steps for each projection interval:
  - estimate the number of people still alive at the start of next interval (**mortality**)
  - compute number of births, aggregate them across groups and compute how many survive to the next interval (**fertility** and **infant mortality**)
  - add immigrants and subtract emigrants, compute births from immigrants, project number of migrants and their newborns in the next interval (**migration**, **fertility** and **mortality**)

CCM - step 1

Estimate the number still alive at the start of next interval (**mortality**)

- if groups are only age and sex, only need single decrement life tables by sex
- use survivorship ratios to compute survivors in the next period, assigned to the next age group (because age and time intervals congruent)
- if more subgroups, need to take into account transitions to different groups - multistate projections (see, e.g., Rogers 1995)

Closed female population

Let's start from a female population, broken down by 5 year age-groups, closed to migration. We need:

- ${}_5N_x^F(t)$, the number of females in age group x to $x+5$ at time t
- ${}_5L_x^F$, the number of person-years lived by females between ages x and $x+5$

Let's drop the left subscript to ease notation

We can compute survivorship ratios s_x^F and the projected population in each age group (except the youngest and oldest):

$$\begin{aligned} s_x^F &= \frac{L_{x+5}^F}{L_x^F} \\ N_x^F(t+5) &= N_{x-5}^F(t)\, s_{x-5}^F \end{aligned} \tag{1}$$

Closed female population: Exercise

Exercise

Open your R session. Load the `dta.swe.1993.Rdata` dataset, and compute the projected population by age groups.

(Do not worry about the first and last age groups, we will adjust them later)

Reminder: first compute $s_x^F = \frac{L_{x+5}^F}{L_x^F}$, and then use the formula

$$N_x^F(t+5) = N_{x-5}^F(t)\, s_{x-5}^F$$

Hint: experiment with the `lead` and `lag` functions of `tidyverse`

One possible solution

One possible solution

```
# A tibble: 6 × 6
    Age AgeGroup    NFx    LFx   sFx    NFx5
  <dbl> <fct>     <dbl>  <dbl> <dbl>   <dbl>
1     0 00-04    293395 497487 0.999     NA
2     5 05-09    248369 497138 1.00  293189.
3    10 10-14    240012 496901 0.999 248251.
4    15 15-19    261346 496531 0.999 239833.
5    20 20-24    285209 495902 0.999 261015.
6    25 25-29    314388 495168 0.998 284787.
```

```
# A tibble: 6 × 6
    Age AgeGroup    NFx    LFx    sFx    NFx5
  <dbl> <fct>     <dbl>  <dbl>  <dbl>   <dbl>
1    60 60-64    215141 457852  0.953 205109.
2    65 65-69    221764 436153  0.923 204945.
3    70 70-74    223506 402775  0.870 204793.
4    75 75-79    183654 350358  0.775 194419.
5    80 80-84    141990 271512  1.07  142324.
6    85 85+      112424 291707 NA     152551.
```

The last age group

For the open-ended age group, we should also consider the number of survivors that were already present in the group at the beginning of the interval. Hence, combining two age groups:

$$N^F_{85+}(t+5) = \left(N^F_{80}(t) \frac{L^F_{85+}}{L^F_{80}} \right) + \left(N^F_{85+}(t) \frac{T^F_{90+}}{T^F_{85}} \right)$$

This requires the open-ended age group in the life table to start at an age (at least) five years older than the population. If we do not have this information, we should use:

$$N^F_{85+}(t+5) = \left(N^F_{80}(t) + N^F_{85+}(t) \right) \frac{T^F_{85+}}{T^F_{80}}$$

We should thus adjust $s^F_{80} = \frac{T^F_{85+}}{T^F_{80}} = \frac{L^F_{85+}}{L^F_{80}+L^F_{85+}}$

The last age group: Exercise

Exercise

Adjust the last age group of your projection

Reminder:

$$s_{80}^{F} = \frac{L_{85+}^{F}}{L_{80}^{F} + L_{85+}^{F}}$$

$$N_{85+}^{F}(t+5) = \left(N_{80}^{F}(t) + N_{85+}^{F}(t)\right) s_{80}^{F}$$

Hint: use the `ifelse` function inside `mutate`

One possible solution

One possible solution

Before adjustment:

```
# A tibble: 3 × 6
    Age AgeGroup    NFx    LFx    sFx    NFx5
  <dbl> <fct>     <dbl>  <dbl>  <dbl>   <dbl>
1    75 75-79    183654 350358  0.775 194419.
2    80 80-84    141990 271512  1.07  142324.
3    85 85+      112424 291707 NA     152551.
```

After adjustment:

```
# A tibble: 3 × 6
    Age AgeGroup    NFx    LFx    sFx    NFx5
  <dbl> <fct>     <dbl>  <dbl>  <dbl>   <dbl>
1    75 75-79    183654 350358  0.775 194419.
2    80 80-84    141990 271512  0.518 142324.
3    85 85+      112424 291707 NA     131768.
```

CCM - step 2

Compute number of births, aggregate them across groups and compute how many survive to the next interval (**fertility** and **infant mortality**)

- ideally: model explicitly couple and union creation and dissolution, births are a by-product of these
- in practice: “female-dominant” approach, births produced by women only
- apply fertility rates to women, disaggregate by sex using sex-ratio at birth
- if more subgroups, additional layer of difficulty - typically assume birth belonging to same group as mother

The first age group I

Let F_x denote the fertility rate at age x. During the projection interval, the number of births to women aged x to $x+5$ is then:

$$B_x[t, t+5] = F_x \cdot 5 \left[\frac{N_x^F(t) + N_x^F(t+5)}{2} \right]$$

where the orange term is an approximation of person-years lived at ages x to $x+5$

Using Equation 1, the total number of births during the period, $B[t, t+5]$ can be written as:

$$B[t, t+5] = \sum_x \frac{5}{2} F_x \left(N_x^F(t) + N_{x-5}^F(t) \frac{L_x^F}{L_{x-5}^F} \right)$$

For now, we are interested in female births $\Rightarrow$ use sex-ratio at birth (SRB):

$$B^F[t, t+5] = \frac{1}{1+SRB} B[t, t+5]$$

The first age group II

- number of females in the first age group is obtained by surviving female births through time $t+5$
- assuming that births are distributed evenly during projection period:

$$N_0^F(t+5) = B^F[t, t+5]\frac{L_0^F}{5\,\ell_0} \qquad (2)$$

  where ℓ_0 is the life-table radix
- equivalently, we can rewrite Equation 2 as:

$$N_0^F(t+5) = \sum_x N_x^F(t) b_x^F$$

where $b_x^F = \frac{1}{1+SRB}\frac{L_0^F}{2\ell_0}\left(F_x + s_x^F\, F_{x+5}\right)$

The first age group: Exercise

Exercise

Adjust the first age group of your projection, assuming a sex-ratio at birth (SRB) of 1.05.

Two possible solutions: we can compute

$$b_x^F = \frac{1}{1+SRB}\frac{L_0^F}{2\ell_0}\left(F_x + s_x^F\, F_{x+5}\right),$$
$$N_0^F(t+5) = \sum_x N_x^F(t) b_x^F$$

Alternatively, we can use

$$B_x[t,t+5] = F_x\, 5\left[\frac{N_x^F(t)+N_x^F(t+5)}{2}\right]$$
$$N_0^F(t+5) = \frac{1}{1+SRB}\frac{L_0^F}{5\,\ell_0}\sum_x B_x[t,t+5]$$

One possible solution

One possible solution

Before adjustment:

```
# A tibble: 1 × 6
    Age AgeGroup    NFx    LFx   sFx  NFx5
  <dbl> <fct>     <dbl>  <dbl> <dbl> <dbl>
1     0 00-04    293395 497487 0.999    NA
```

```
[1] NA
```

After first adjustment:

```
# A tibble: 1 × 6
    Age AgeGroup    NFx    LFx   sFx    NFx5
  <dbl> <fct>     <dbl>  <dbl> <dbl>   <dbl>
1     0 00-04    293395 497487 0.999 293574.
```

```
[1] 293573.8
```

Alternative formula:

```
[1] 293573.8
```

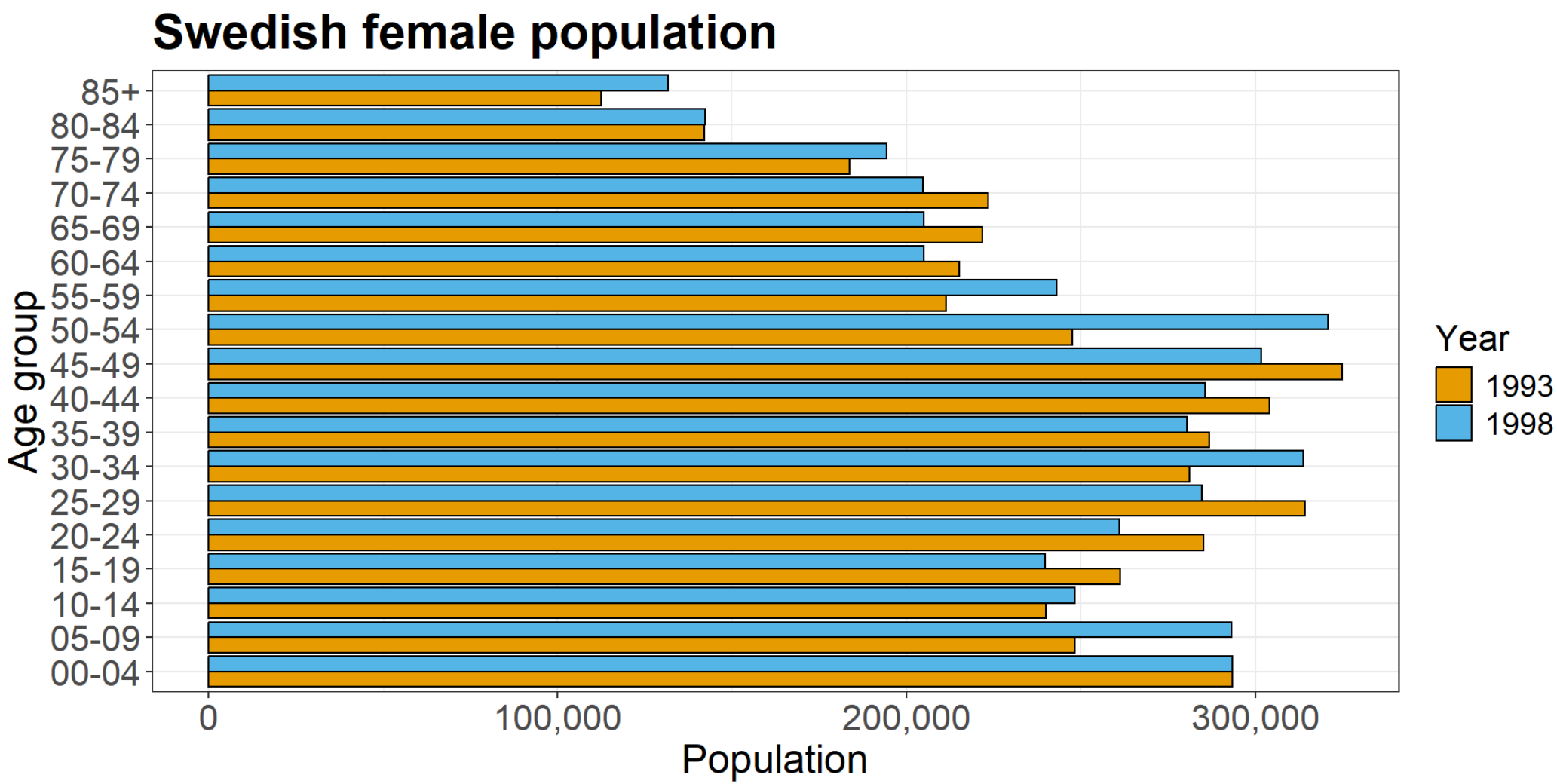
Plotting results with pyramids

```r
## useful packages
library(scales)

## long data
dta.swe.l <- dta.swe %>%
    select(AgeGroup,NFx,NFx5) %>%
    rename('1993'=NFx,'1998'=NFx5) %>%
    pivot_longer(-AgeGroup,names_to = "year",values_to = "populatic

## plotting
ggplot(dta.swe.l,aes(x=AgeGroup,y=population,fill=year)) +
    geom_bar(stat = "identity",position = "dodge",color = "black")
    coord_flip() +
    theme_bw() +
    ggtitle("Swedish female population") +
    scale_y_continuous(labels = scales::comma) +
    scale_fill_manual(values=c("#E69F00", "#56B4E9"))
```

Plotting results with pyramids



Two-sex closed population

The male population can be projected with the same formulas and using male survivorship ratios

$$s_x^M = \frac{L_{x+5}^M}{L_x^M} \quad ; \quad s_{80}^M = \frac{T_{85+}^M}{T_{80}^M}$$

$$N_x^M(t+5) = N_{x-5}^M(t)\, s_{x-5}^M$$

$$N_{85+}^M(t+5) = \left(N_{80}^M(t) + N_{85+}^M(t)\right) s_{80}^M$$

$$b_x^M = \frac{SRB}{1+SRB}\frac{L_0^M}{2\ell_0}\left(F_x + s_x^F\, F_{x+5}\right)$$

$$N_0^M(t+5) = \sum_x N_x^F(t) b_x^M$$

Two-sex closed population: Exercise

Exercise

Now compute the male projected population by age groups, adjusting the first and last age groups. Then save your data, as this will be useful for tomorrow's lecture.

$$s_x^M = \frac{L_{x+5}^M}{L_x^M} \quad ; \quad s_{80}^M = \frac{T_{85+}^M}{T_{80}^M}$$

$$N_x^M(t+5) = N_{x-5}^M(t)\, s_{x-5}^M$$

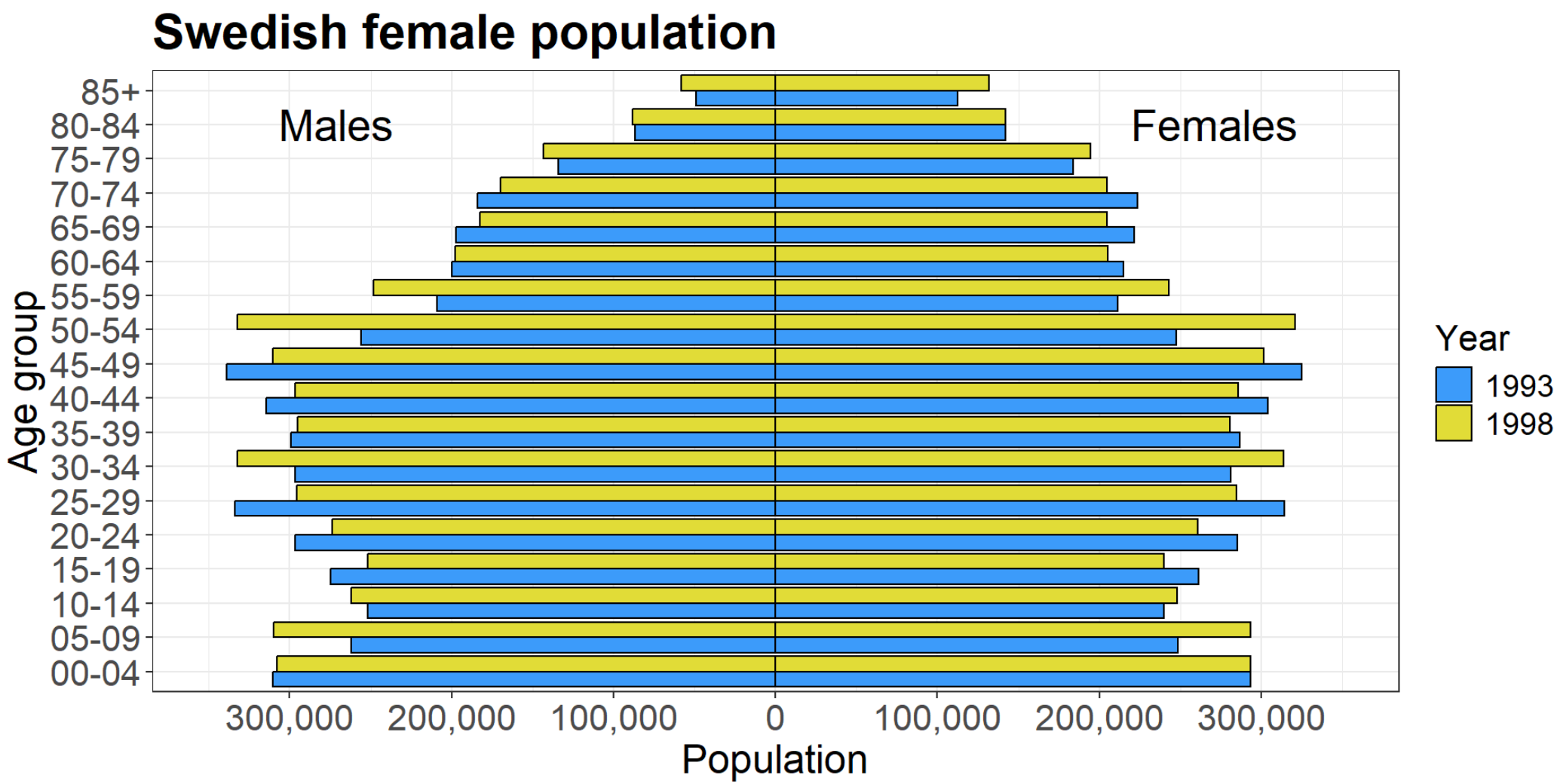
$$N_{85+}^M(t+5) = \left(N_{80}^M(t) + N_{85+}^M(t)\right) s_{80}^M$$

$$b_x^M = \frac{SRB}{1+SRB}\,\frac{L_0^M}{2\ell_0}\left(F_x + s_x^F\, F_{x+5}\right)$$

$$N_0^M(t+5) = \sum_x N_x^F(t) b_x^M$$

One possible solution

One possible solution



Some final remarks

- cohort component method as the most employed model for population projections (when data available)
- takes into account age composition of populations
- projection of age groups rather straightforward, except for the youngest and the eldest
- projected population from one interval becomes the baseline for next projection
- several projections can become cumbersome if done one at a time

$\Rightarrow$ matrix algebra can help us to speed this up (tomorrow's lecture)

Day 1 assignment

Assignment

1. Choose one country of interest. Download and compare its population projections from at least three data providers (see Global population projections). What are the main differences between the projections, in terms of both total size and age-sex composition?

2. Take the population for any country (for example, use data from the HMD), as well as data for person-years and fertility rates (for example, take data from HMD and HFD). Specifically, take these data 5 years before the last available date (e.g.~if the last available year is 2024, take the data in 2019). Divide the population into 5-year age groups and project the population 5-year ahead for each sex separately. Plot and compare your results with the observed population in the last available data (e.g. in 2024). Are your projections close to the observed population? At what ages do you see larger differences? Briefly provide a motivation for this discrepancy.

References

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